

STAT 215A Lab 2

2026-10-02

Introduction

The Harvard Dialect Survey (Vaux et al.) asked tens of thousands of respondents across the United States a list of questions about everyday word choice, in the hope of mapping regional variation in American English vocabulary in the way that traditional dialectologists mapped regional variation in pronunciation. This lab works with a cleaned excerpt of that survey: 47,471 respondents' answers to 67 "lexical" questions (word-choice questions, as opposed to the survey's phonetic questions), together with each respondent's self-reported city/state and a geocoded latitude and longitude.

Aggregate, quantitative analysis of exactly this kind of data has a long history in dialectology. Nerbonne and Kretzschmar's two review papers make the case that pooling many small, individually noisy differences together is what lets quantitative methods recover real dialect regions and describe how sharp or gradual the boundaries between them actually are (Nerbonne and Kretzschmar, 2003, 2006; full citations in the Bibliography). That is the strategy of this lab. I first look at two individual survey questions by hand, then recode every respondent's 67 answers into a single binary vector and use dimension reduction and clustering to see whether "aggregate" dialect regions emerge from the data, whether they match the broad regions described in the dialectology literature (roughly a North / Midland / South split, refined by coastal and inland differences), and how stable those regions are to resampling and to the clustering algorithm's random initialization.

The Data

`lingData.txt` has one row per respondent ($n = 47,471$) and columns `ID`, `CITY`, `STATE`, `ZIP`, the 67 lexical questions `Q050-Q121`, and a geocoded `lat/long`. Each `Q` column holds an integer code for the option the respondent chose (1, 2, 3, ...), with 0 meaning "no response"; the text of each numbered option for each question is given in a companion file, `question_data.RData`. `lingLocation.txt` aggregates the same 67 questions into $1^\circ \times 1^\circ$ latitude/longitude grid cells, with each cell's answers already summed into one-hot counts. I use it only as a sanity check on my aggregation and otherwise work with the respondent-level `lingData.txt`.

Data Cleaning

Because this is a self-reported survey, several cleaning decisions were necessary.

- **STATE is messy free text.** It takes 92 distinct raw values in the data, and many of them (e.g. PL, 00, IS, !L) are not valid two-letter USPS codes. I decided so that any `STATE`-based summary below is restricted to the subset of rows whose `STATE` is a valid USPS code.
- **lat/long is missing for 1,020 respondents (2.1%)** whose city/state could not be geocoded. I drop these respondents only from analyses that require a location (maps) but kept them everywhere else.
- **206 further geocoded respondents (0.4%) fall outside the continental United States** – true Alaska/Hawaii addresses, plus a handful of entries whose self-reported mainland city/state (e.g. "SanDiego, CA") was evidently geocoded to a Hawaiian ZIP code. I exclude these from maps of

the continental United States, since the dialect regions I discuss below are mainland regions, but keep them in the non-geographic clustering analysis.

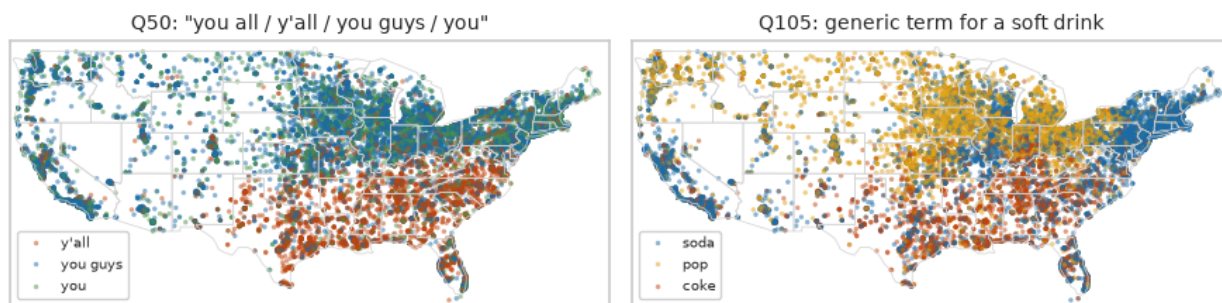
- **A Q value of 0 means “did not answer that question,”** As a judgement call, I choose to treat it as missing rather than as its own category.
- **1,040 respondents (2.2%) did not answer any of the 67 lexical questions at all** so I choose to drop them entirely.

Exploratory Data Analysis

I asked AI to list a few interesting questions for me and I picked: **Q50**, “What word(s) do you use to address a group of two or more people?” (answers include *you*, *you guys*, *y’all*, *you all*, *yous/youse*, *you’uns*, *yins*), and **Q105**, “What is your generic term for a sweetened carbonated beverage?” (*soda*, *pop*, *coke*, *tonic*, *soft drink*, ...). Both are among the most famous individual dialect maps of American English, which makes them a useful place to start with

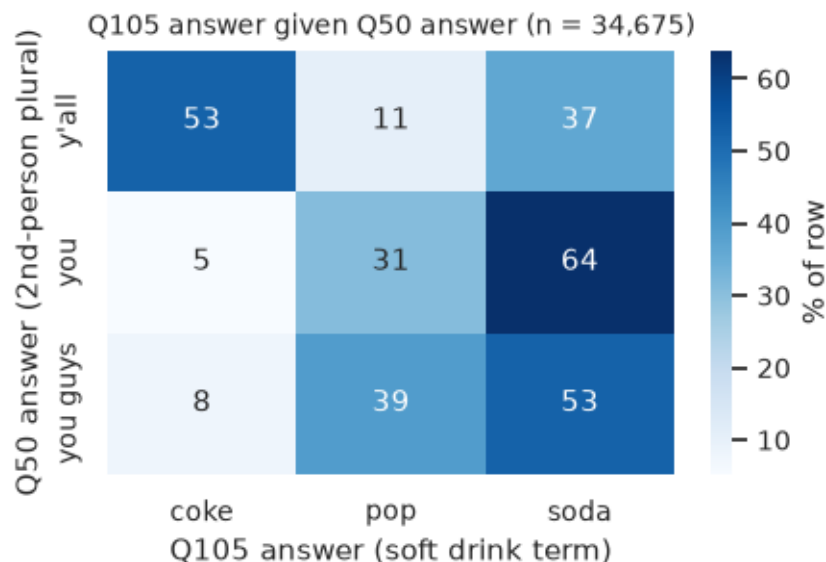
Mapping the two most regionally distinctive answers to each question shows that both individually define clear (and partly different) geographic groups.

Two classic lexical dialect maps



For Q50, *y’all* is overwhelmingly a Southern answer, *you guys* dominates the West Coast, the industrial Midwest, and much of the interior Northeast, and the plain *you* is a minority answer concentrated in the Northeast and Mid-Atlantic. For Q105, *soda* dominates the Northeast and the West Coast, *pop* dominates a broad band of the Midwest, Great Lakes, and Mountain West, and *coke* is used as a generic term for any soft drink across almost the entire South. Both questions individually recover a recognizable geographic pattern that is broadly consistent with the North/Midland/South regions discussed in the dialectology literature, even though neither question mentions geography at all.

Since both questions seem to be tracking the same underlying “which region is this person from” variable, a natural next question is whether the answer to one predicts the answer to the other, even with no geographic information at all.



The two questions are strongly associated: among respondents who say *y'all*, 52.7% also say *coke* and only 10.7% say *pop*, while among respondents who say the plain *you*, only 5.3% say *coke* and 63.9% say *soda*; *you guys* respondents sit in between, with a plurality (52.7%) saying *soda* and a substantial 39.3% saying *pop*. Knowing the answer to one question is clearly informative about the other, even though the two questions have nothing to do with each other semantically. That is the idea I pursue for the rest of the report: if two individually noisy lexical variables can already be partially explained by a shared regional signal, pooling information across all 67 questions at once should recover that regional structure far more clearly than any single question or pair of questions can.

Dimension Reduction

To use all 67 questions at once I first need a numeric encoding of each respondent. The raw **Q** columns are *nominal* categorical codes: option 4 of Q50 (*you guys*) and option 9 (*y'all*) are just labels, and the difference “9 minus 4 = 5” means nothing linguistically. Treating these codes as ordinary numbers and feeding them straight into PCA (or any other Euclidean-distance-based method) silently assumes an order and a scale that isn’t there, and different questions have very different numbers of options (from 2 to 21), so a question with more options gets an arbitrarily larger numeric range and can dominate the variance for no substantive reason. I can see this failure mode directly: running PCA on the raw **Q** columns gives a first “principal component” that explains 23% of the variance, but 0.95 of its loading sits on a single question, Q59 (which happens to have 21 answer options, the most of any question in the data) – the correlation between a question’s raw-code variance and its number of answer options is 0.80. The first PC is not telling me about regional dialect variation at all; it is telling me which question happened to have the widest arbitrary numbering scheme.

Instead I follow the encoding suggested in the instructions and turn each question into a set of binary indicators, one per answer option (a respondent who chose option k of a question gets a 1 in that question’s k -th indicator column and a 0 everywhere else for that question; a respondent who did not answer the question gets all zeros for it). Summed over the 67 lexical questions this produces $p = 468$ binary columns per respondent, matching the instructions.

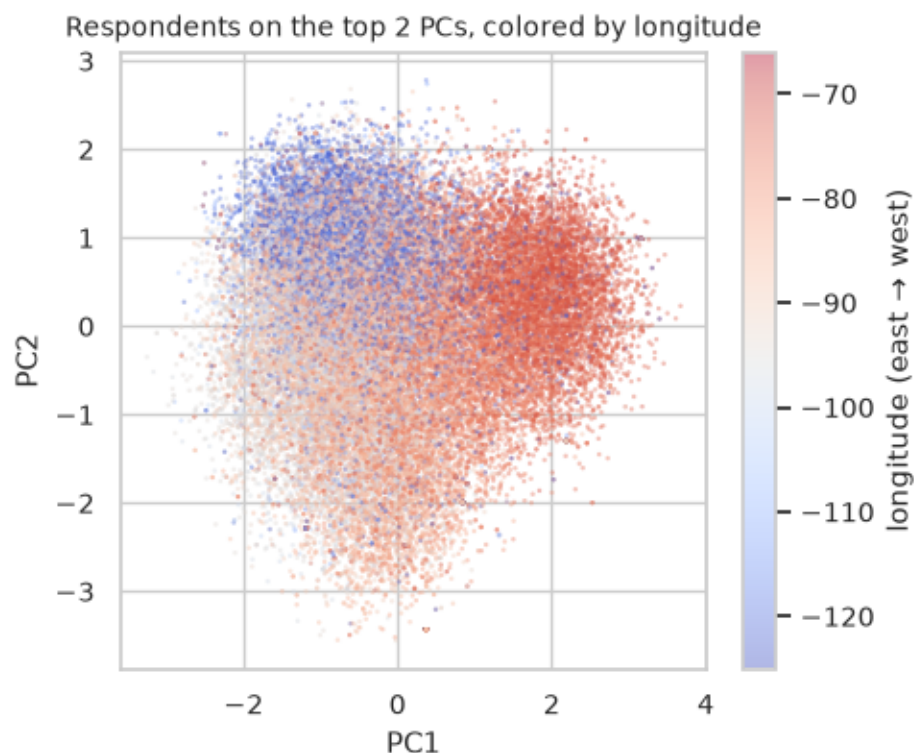
Scaling

`sklearn`’s PCA centers the data automatically. The remaining choice is whether to also scale: many of the 468 one-hot indicator columns correspond to rare answers chosen by a tiny fraction of respondents, while others are chosen by close to half the sample, so standardizing every column to unit variance (as in a correlation-matrix

PCA) would treat them as equally informative. The next cell checks what that actually does to the columns and to the resulting components.

Scaling is a bad idea here. The rarest column (Q067_10, chosen by just 4 of 46,431 respondents) has such tiny variance that standardizing it to unit variance inflates it by roughly 108x, versus only 2x for the most evenly split column. The data demonstrates that the top five components explain only about half as much variance scaled (7.2%) as unscaled (14.6%), because scaling spreads the signal across more, noisier components instead of concentrating it in a few interpretable ones. I therefore use centered, unscaled PCA throughout the rest of the report.

Even with 50 components I only reach 55.7% of the total variance (33.0% with 20 components). This is expected for high-dimensional, sparse survey data where each of the 67 questions contributes its own largely independent axis of variation, and it means no single pair of PCs should be expected to show all of the interesting structure at once.



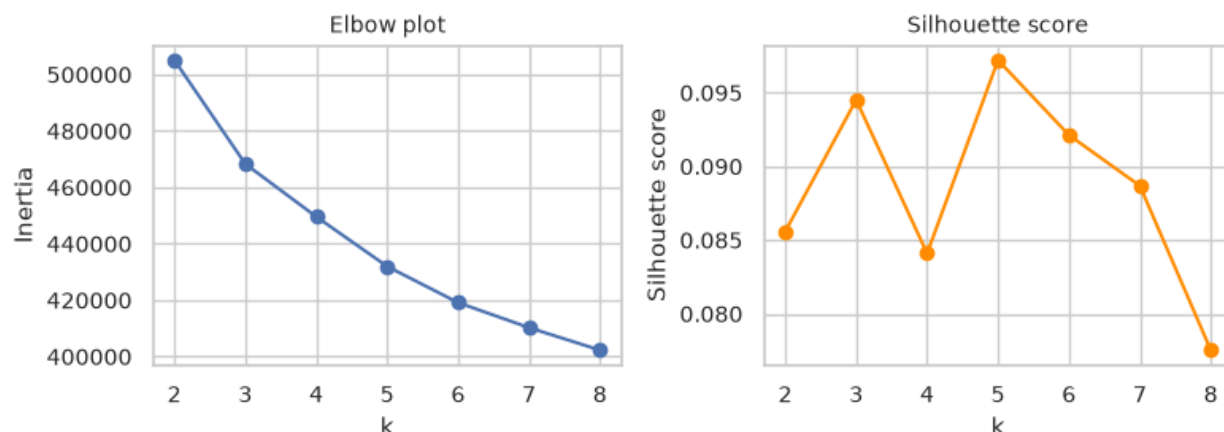
PC1 and PC2 alone show a diffuse cloud with only a weak color gradient which is consistent with the low variance explained by any single component. If I stopped here I would conclude “I don’t see much,” exactly the situation the instructions anticipate, however changing the projection helps. Coloring by longitude reveals a real, if noisy, East-West gradient along PC1 once I look for it, and (as I show next) using more than 2 PCs as the input to a clustering algorithm recovers much cleaner regional structure than either PC1 or PC2 shows by itself. I use the top 20 PCs (33.0% of variance) as the working low-dimensional representation for the clustering analysis below, which balances keeping most of the structure PCA finds against not re-introducing the noise in components far down the spectrum.

Clustering

I cluster respondents in the 20-PC space described above using two different algorithms: *k*-means, and Ward’s hierarchical (agglomerative) clustering.

Choosing the number of clusters

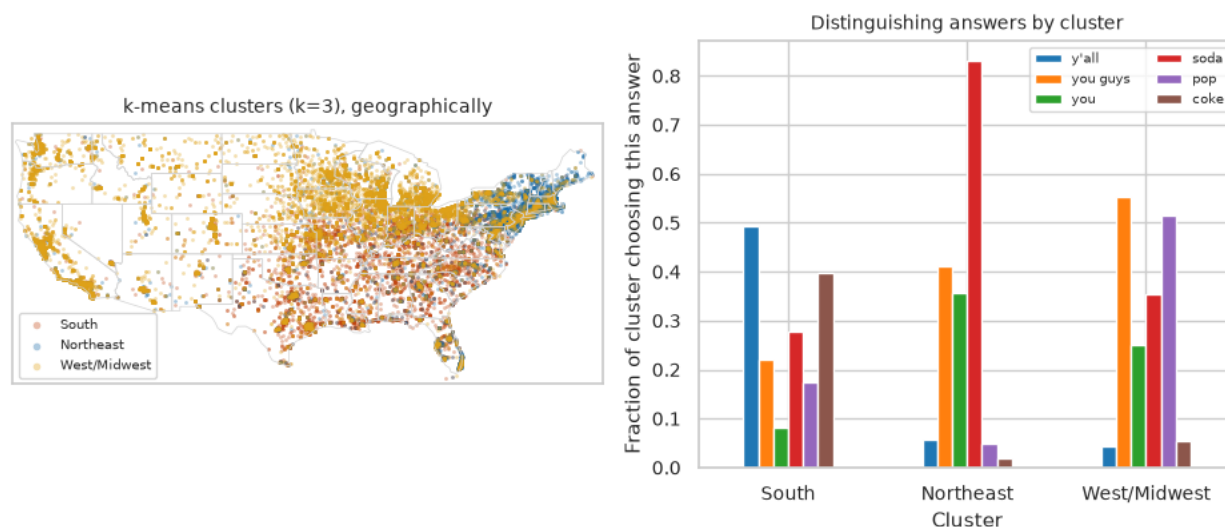
I ran k -means for $k = 2, \dots, 8$ and evaluated both the within-cluster sum of squares (inertia) and the silhouette score (on a random subsample of 5,000 respondents, for speed).



Neither curve shows a single sharp elbow: this is itself informative. I choose $k = 3$ for two reasons beyond the silhouette score. First, it is more interpretable and matches the coarse North/Midland-or-West/South structure suggested by the two-question EDA above. Second, inspecting $k = 5$ shows why more clusters is not obviously better: one of its five clusters contains only 636 respondents, and every single one of them (100%) answered *y'all* to Q50. This is a useful illustration of a general disadvantage of k -means: because it only minimizes Euclidean distance to k cluster centers, it has no way to distinguish “a genuinely new subpopulation” from “an unusually extreme tail of an existing one,” and will produce whichever partition is requested regardless.

What the three clusters look like

Mapping the three k -means clusters and looking at each cluster’s most distinctive answers to Q50 and Q105 shows that they align closely with familiar dialect geography.



The **South** cluster’s top states by respondent count are TX, MO, VA, KS, and OH; 49.4% of its members say *y'all* and 39.7% say *coke*. The **West/Midwest** cluster is the largest and most geographically spread out (top states CA, IL, MI, WA, WI); 55.3% say *you guys* and 51.6% say *pop* which aligns with the general American pattern. The **Northeast** cluster’s top states are NY, PA, MA, NJ, and CT; 83.1% say *soda*, easily its most

distinctive feature, and it has the highest rate (35.8%) of the plain *you*. All of this lines up closely with the individual Q50/Q105 maps from the Exploratory Data Analysis section, but now derived automatically from all 67 questions at once rather than hand-picked from two.

A second clustering method: hierarchical clustering

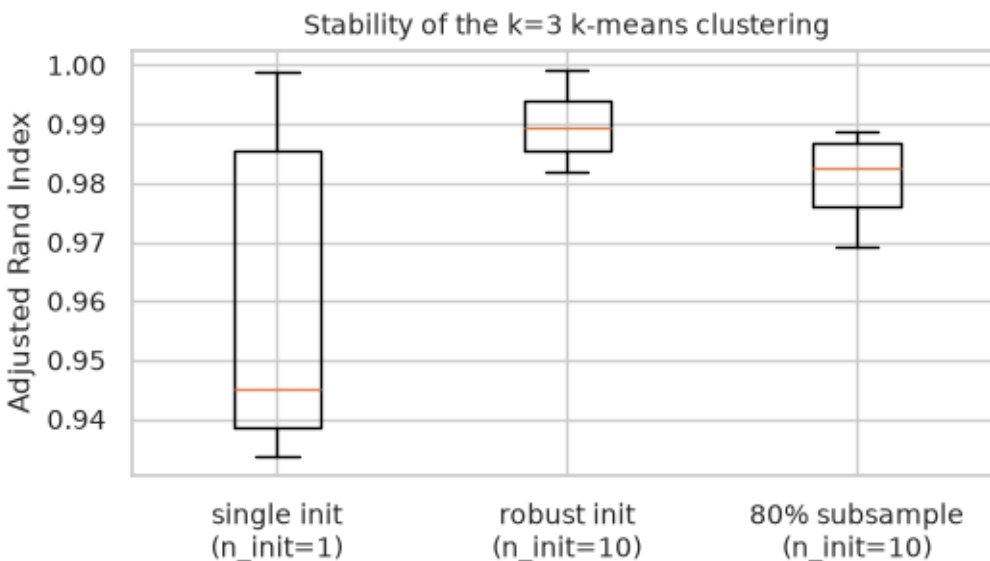
As a second method I ran Ward’s hierarchical clustering, which builds a full dendrogram by repeatedly merging the two clusters whose merger increases within-cluster variance the least. Hierarchical clustering scales poorly to $n = 46,431$ points (its distance computation is $O(n^2)$), so I fit it on a random subsample of 3,000 respondents and cut the resulting dendrogram at 3 clusters, then compared the result to k -means fit on the same subsample.

The two methods agree far more than chance (Adjusted Rand Index 0.46 – 1.0 would be perfect agreement, 0.0 is what two random partitions would give) but clearly do not agree perfectly. This moderate-but-incomplete agreement, together with the lack of a sharp elbow above, is itself evidence for one of the questions the instructions raise: the boundaries between these three regions look more like a continuum than three cleanly separated islands, which matches how dialectologists usually describe American English regions.

Each method has its own advantages and disadvantages for this application. k -means scales to the full $n = 46,431$ respondents, is fast, and directly optimizes an interpretable objective, but it must be told k in advance. Ward’s method does not require choosing k up front and its merge history is a useful diagnostic in its own right, but it does not scale to the full dataset and, because it greedily merges from the bottom up, an early bad merge cannot be undone later.

Stability of findings to perturbation

I focus on the k -means, $k = 3$ result and ask how much it changes under two kinds of perturbation: (1) re-running k -means from different random starting points, and (2) re-running it on random 80% subsamples of the data. In both cases I measure agreement between two clusterings with the Adjusted Rand Index (ARI), which is 1.0 for identical partitions and about 0.0 for two random partitions.



The clustering is quite stable. A single, un-restarted k -means run already agrees with another single run 0.93-0.99 of the time (mean ARI 0.96); using scikit-learn’s default of 10 internal restarts per fit (as I did throughout the report) tightens this further to a mean of 0.99. Clustering random 80% subsamples of the data and comparing labels on the overlapping respondents gives a similarly high and tight distribution of ARIs (mean 0.98, sd 0.006). In other words, the three-region structure is not an artifact of a lucky random

initialization or of the particular 46,431 respondents I happened to have. After the stability check, it reappears reliably under both kinds of perturbation I tried, which gives me much more confidence in the geographic story above than the (fairly low) silhouette scores alone would suggest. This is consistent with my reading of the South/Northeast/West-Midwest split as a real, broad regional grouping and not noise.

Conclusion

Running through the three realms (data, algorithms and analysis, and future data) is a useful way to check how much weight these results can actually carry. Start with the data itself: this is a convenience sample. People opted into a web survey, so respondents skew toward internet users willing to sit through a dialect quiz, and the geocoding only places someone at their ZIP code's centroid, not their real address. The 92 distinct raw `STATE` values documented in Data Cleaning are a reminder that some of it is simply typed wrong. None of that invalidates the analysis, but it does mean the clusters describe this particular, imperfect sample rather than the U.S. population at large.

That caveat carries over to the algorithms and analysis: the moderate agreement between k -means and Ward's method, together with the lack of a sharp elbow in either, points toward a real but graded regional gradient rather than three hard-edged boundaries. I'd treat the three clusters as a reasonable summary of that gradient, not as ground truth.

For future data and decisions, I'd treat this dataset as a snapshot. It is useful for describing lexical variation around when the survey ran, but not something I'd want to build a decision on without revisiting it. Language use drifts over time and the sample isn't demographically representative, so I'd be wary of reusing these exact cluster boundaries for something like regional marketing copy, or any other decision that needs current, representative estimates, without re-validating on fresher data first.

The most direct reality check I can think of is to hold out some of the 67 questions, refit the clustering on the rest, and see whether the resulting respondent-level clusters still line up with the held-out questions the way the Q50/Q105 relationship in the Exploratory Data Analysis section suggests they should. I'd also want to rerun the whole pipeline on `lingLocation.txt`'s grid-cell aggregates, as an independent check that the person-level conclusions aren't an artifact of how I cleaned and encoded `lingData.txt`. I didn't get to either of these, but they're the first things I'd try with more time, along with a soft-clustering approach.

Academic Honesty

I value UC Berkeley's standards of academic integrity, as well as academic integrity more broadly. I believe honesty is essential to academic research because research depends on others being able to trust how ideas, analyses, and conclusions were produced. All analyses and conclusions in this report are my own. Whenever I received ideas or consultation from other students, outside resources, or AI tools, I have acknowledged their use in the relevant part of the report.

I used LLM tools in the following ways:

- **Report writing:** I used generative AI to polish grammar and sentence flow in already-written paragraphs in the report. I did not ask it to generate paragraphs from bullet points or rewrite sections wholesale. All edits were reviewed and are reflected in the final wording.
- **Code:**
 - I did not use LLM to generate any code from scratch in this lab.
 - I used AI to help organized so my code follows the guidelines and wrote explanation/documentation for my code. I retyped and verified the suggested code myself before incorporating it.

Bibliography

- J. Nerbonne and W. Kretzschmar Jr. (2003). Introducing computational techniques in dialectometry. *Computers and the Humanities*, 37(3), 245-255.

- J. Nerbonne and W. Kretschmar Jr. (2006). Progress in dialectometry: Toward explanation. *Literary and Linguistic Computing*, 21(4), 387-397.
- Vaux, B. and Golder, S. *The Harvard Dialect Survey*. Harvard University Linguistics Department.